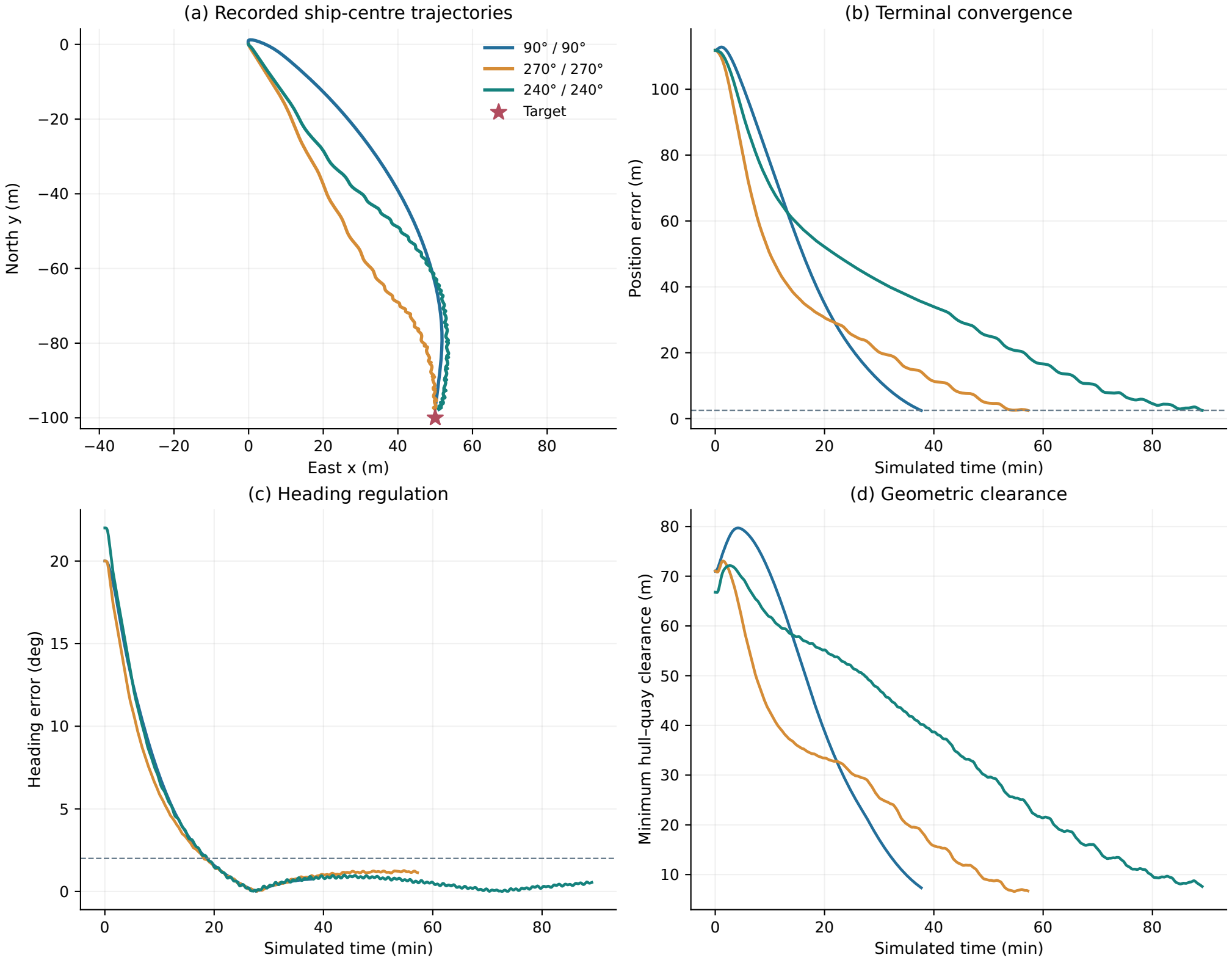
(a) Recorded ship-centre trajectories
90° / 90°
270° / 270°
240° / 240°
Target
North y (m)
East x (m)
(b) Terminal convergence
Position error (m)
Simulated time (min)
(c) Heading regulation
Heading error (deg)
Simulated time (min)
(d) Geometric clearance
Minimum hull–quay clearance (m)
Simulated time (min)